SU
SINAI UNIVERSITY
KANTARA

INT 422
Pattern Recognition

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LECTURE 1

Introduction to Pattern Recognition

Agenda

- Human Perception
- What is a pattern?
- Applications of PR
- Pattern Class
- Inter-class Similarity vs. Intra-class
- Supervised and unsupervised classification
- Pattern Recognition System
- Example





Human Perception

- Humans have developed highly sophisticated skills for sensing their environment and taking actions according to what they observe, e.G.,
  - Recognizing a face
  - Understanding spoken words
  - Reading handwriting
  - Distinguishing fresh food from its smell.
- We would like to give similar capabilities to machines.

Pattern Recognition

- Pattern recognition is the scientific discipline whose goal is the classification of objects into a number of categories or classes.
- "The real power of human thinking is based on recognizing patterns.
- The better computers get at **pattern recognition**, the more human-like they will become." *Ray kurzweil, NY times, nov 24, 2003*
- "The problem of **searching for patterns in data** is a fundamental one and has a long and successful history." Bishop

Pattern Recognition (cont.,)

The act of taking as input sensed data (measurements) and taking an action based on the "**category**" or "**class**" of the pattern.

Pattern Recognition (cont.,)

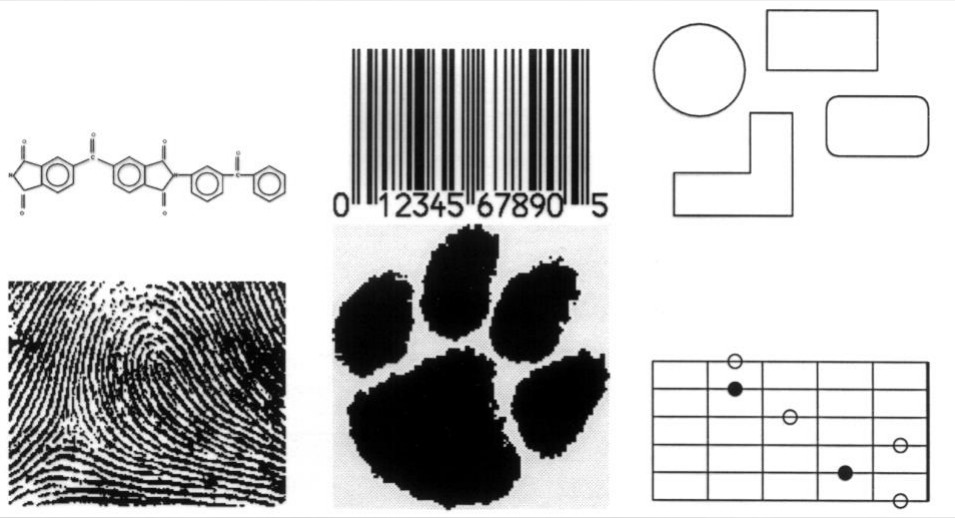
- The study of how machines can observe the environment, learn to distinguish patterns of interest from their background, and make sound and reasonable decisions about the categories of the patterns.
- The assignment of a physical object or event to one of several pre-specified categories

- **Related Terms**
  - ***pattern class*** (or category) is a set of patterns sharing common attributes and usually originating from the same source.
  - ***classifier*** is a machine which performs classification.

What is a Pattern?

- Pattern: opposite to chaos; it is an entity, object, process or event, vaguely defined, that can be given a name or “label”.
- **For *example, a pattern could be***
  - A fingerprint image
  - A handwritten cursive word
  - A human face
  - A speech signal
  - Texture
  - Etc.

What is a Pattern?



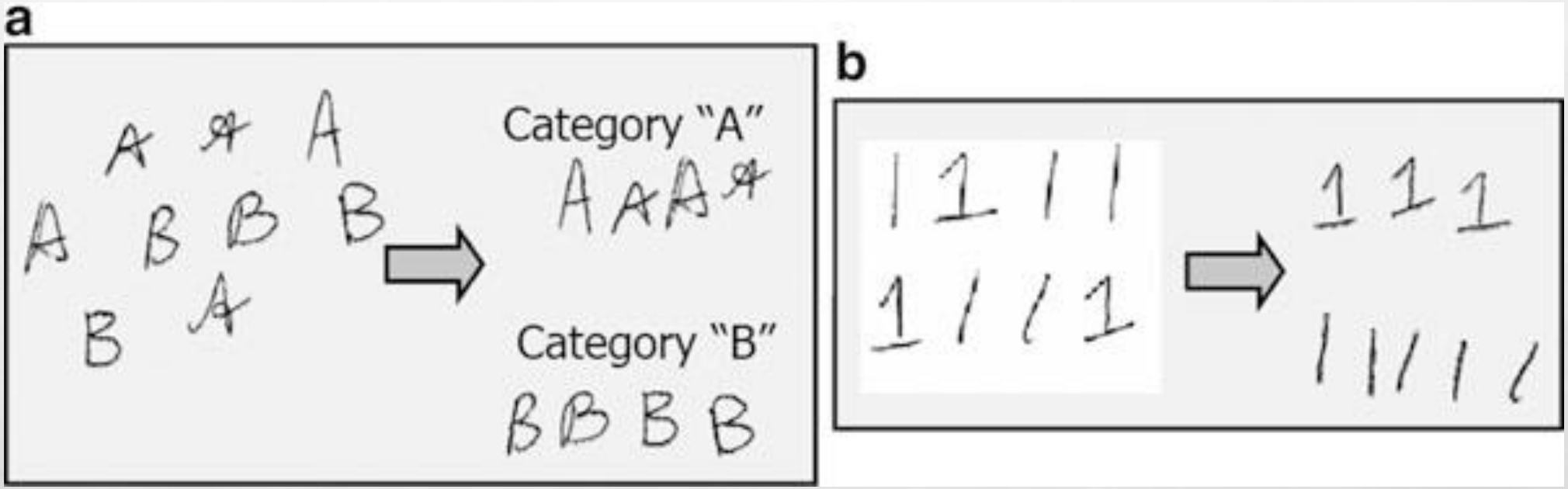




Recognition

- **Identification** of a **pattern** as a member of a **category** (class) we already **know**, or we are **familiar** with
- Classification (known categories) ------→ **Supervised**
- Clustering (learning categories or unknown categories) --→ **Unsupervised**

Recognition



Classification **Clustering**

CAT VS. DOG: 2-CLASS CLASSIFICATION





(SUPERVISED) CLASSIFICATION





Labeled training samples for classifier design
https://www. youtube.com/watch?v=cfj6yaYE86U

Clustering: Unsupervised Classification

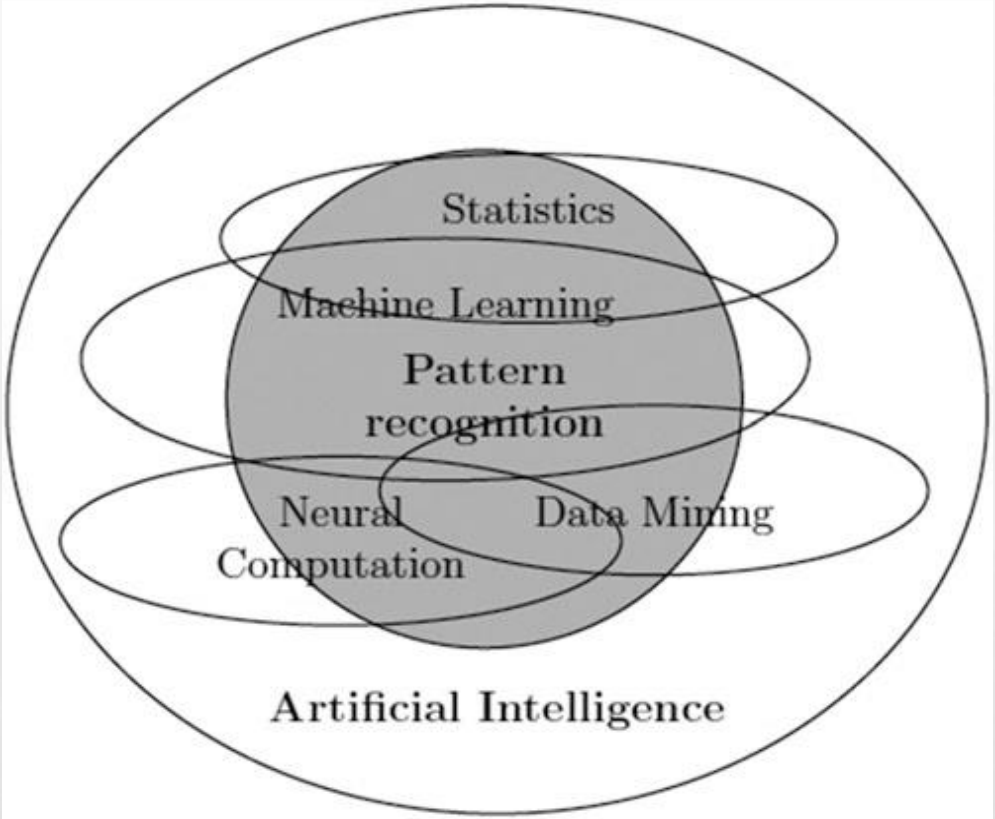


Training samples are unlabeled

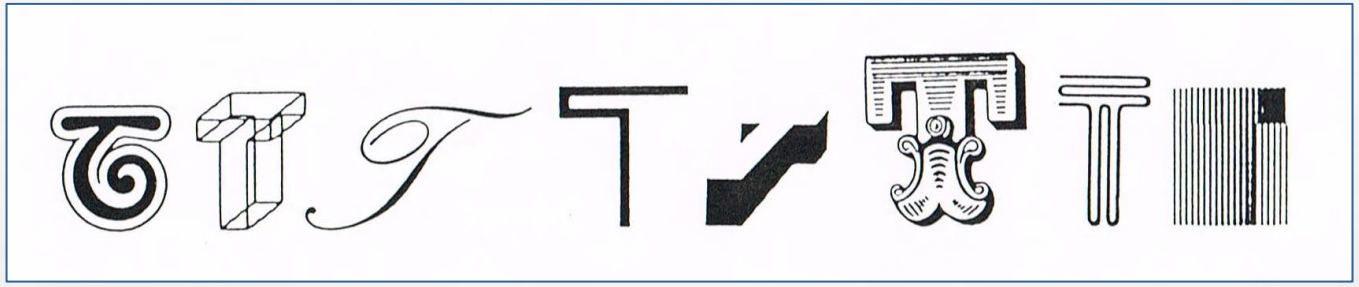
Why Is Pattern Recognition Important?

- Nowadays, pattern recognition serves as a basis for a number of technologies used in everyday life.
- Face recognition can be one of the most common examples of implementing pattern recognition on a complex level.
- Face recognition, as well as other biometrics technologies, have already tremendously influenced the process of identity verification and will continue to influence our society.
- pattern recognition is an irreplaceable analytical tool as well. Complex big data analyses, like stock market prediction, business analytics, or medical diagnostics rely on pattern-recognizing algorithms. Without seamless pattern recognition, drawing meaningful conclusions from large sets of data would be impossible.

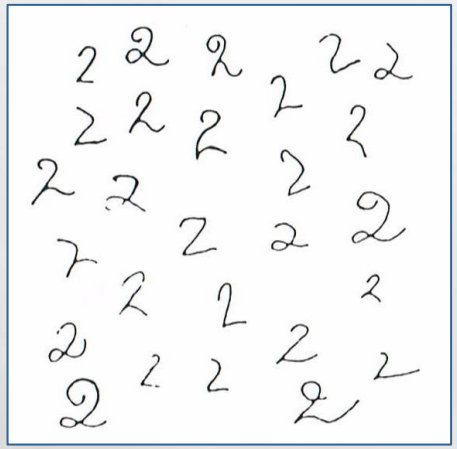
Pattern Recognition and Related Fields

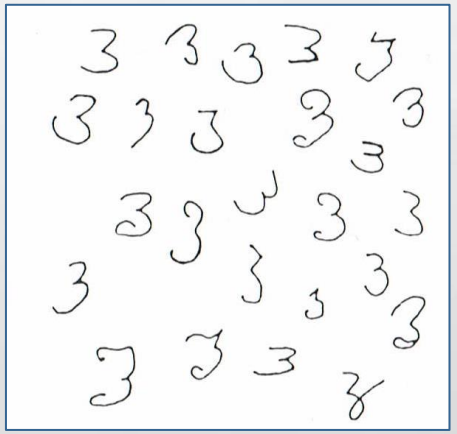


Pattern Class



The letter “T” in different typefaces



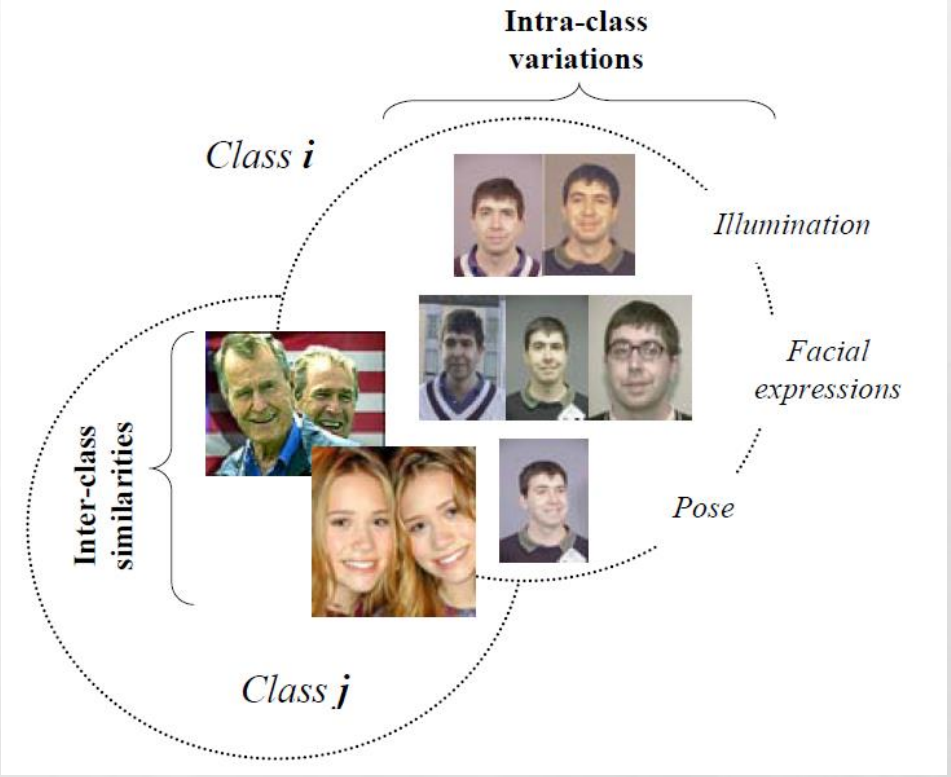


Pattern Class

- A collection of similar (not necessarily identical) objects
- A class is defined by class samples (exemplars, prototypes)
  - Intra-class variability
  - Inter-class similarity
- **How to define similarity?**

Inter-class Similarity vs. Intra-class Variability

- The first one concerns **inter-class similarity** such as twins' classes, fathers and sons' classes. Here, the people have similar appearances which make their discrimination difficult.
- The second and the more important are **the intraclass variations** caused by significant changes in lighting conditions, pose variations (i.E. Three-dimensional head orientation), and facial expressions.



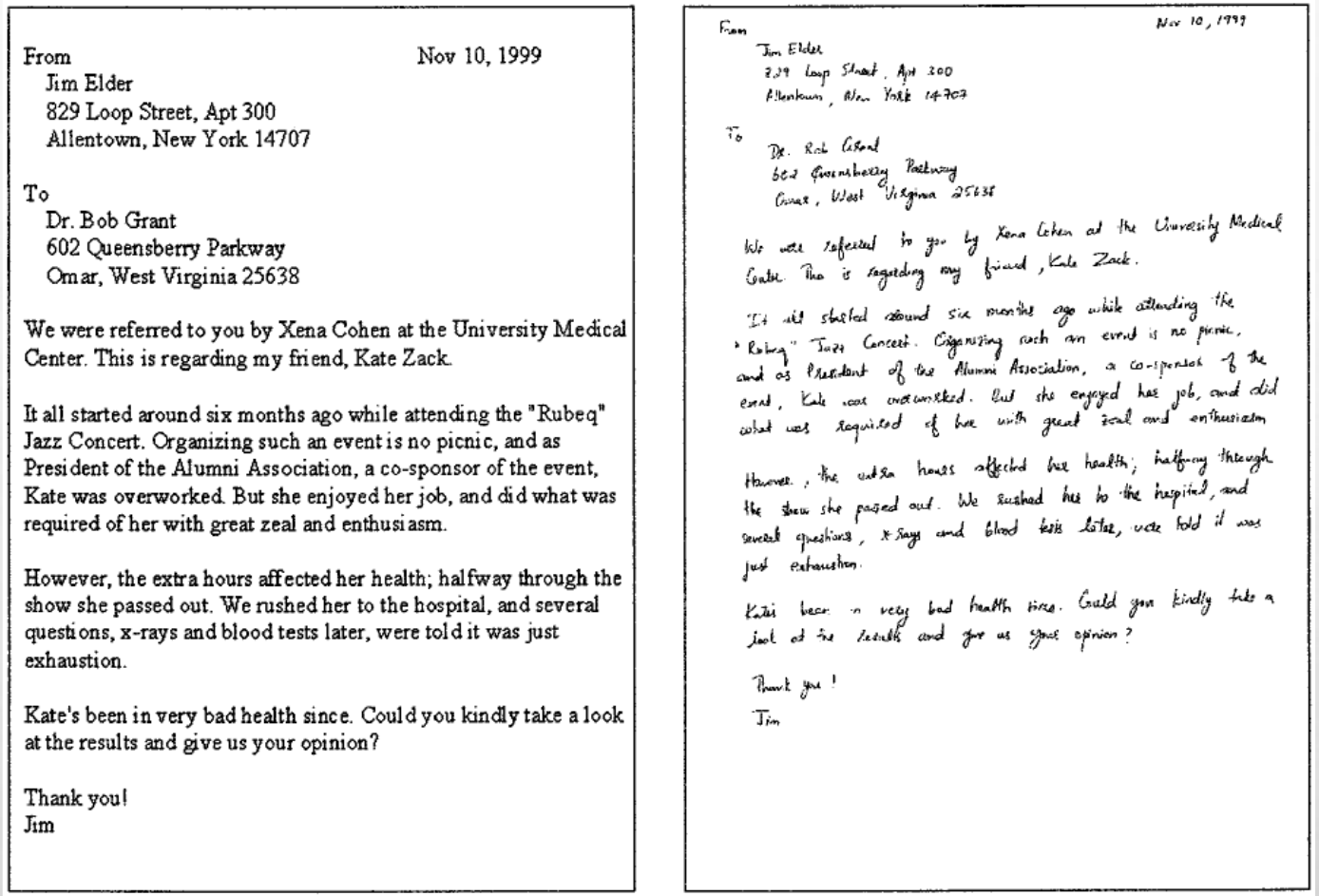
Inter-class similarities and intra-class variations: face recognition problems

Applications of PR



Face recognition needs to be able to handle different expressions, lighting, and occlusions

Applications of PR



From
Jim Elder
829 Loop Street, Apt 300
Allentown, New York 14707

Nov 10, 1999

To
Dr. Bob Grant
602 Queensberry Parkway
Omar, West Virginia 25638

We were referred to you by Xena Cohen at the University Medical Center. This is regarding my friend, Kate Zack.

It all started around six months ago while attending the "Rubeq" Jazz Concert. Organizing such an event is no picnic, and as President of the Alumni Association, a co-sponsor of the event, Kate was overworked. But she enjoyed her job, and did what was required of her with great zeal and enthusiasm.

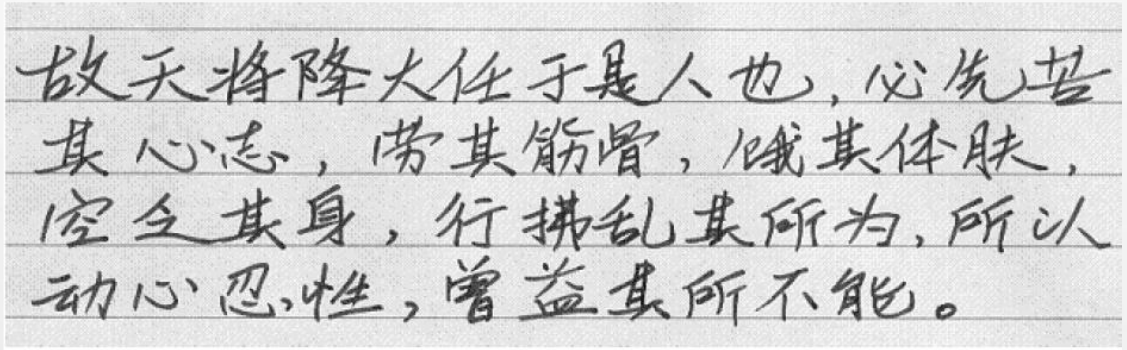
However, the extra hours affected her health; halfway through the show she passed out. We rushed her to the hospital, and several questions, x-rays and blood tests later, were told it was just exhaustion.

Kate's been in very bad health since. Could you kindly take a look at the results and give us your opinion?

Thank you!
Jim

English handwriting recognition

Applications of PR



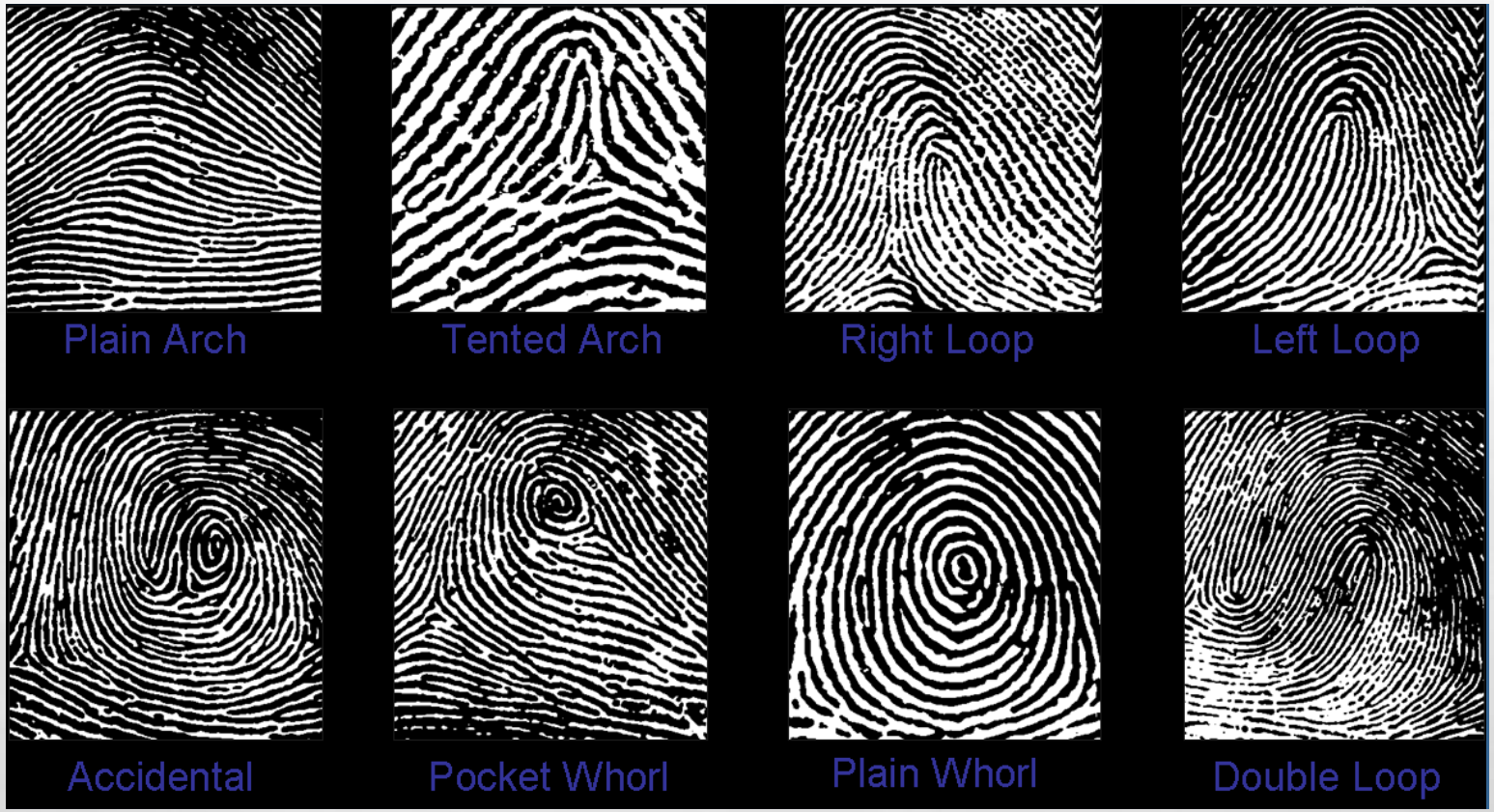
(a) Handwriting

故天将降大任于是人也，必先苦
其心志，劳其筋骨，饿其体肤，
空乏其身，行拂乱其所为，所以
动心忍性，曾益其所不能。

(b) Corresponding Machine Print

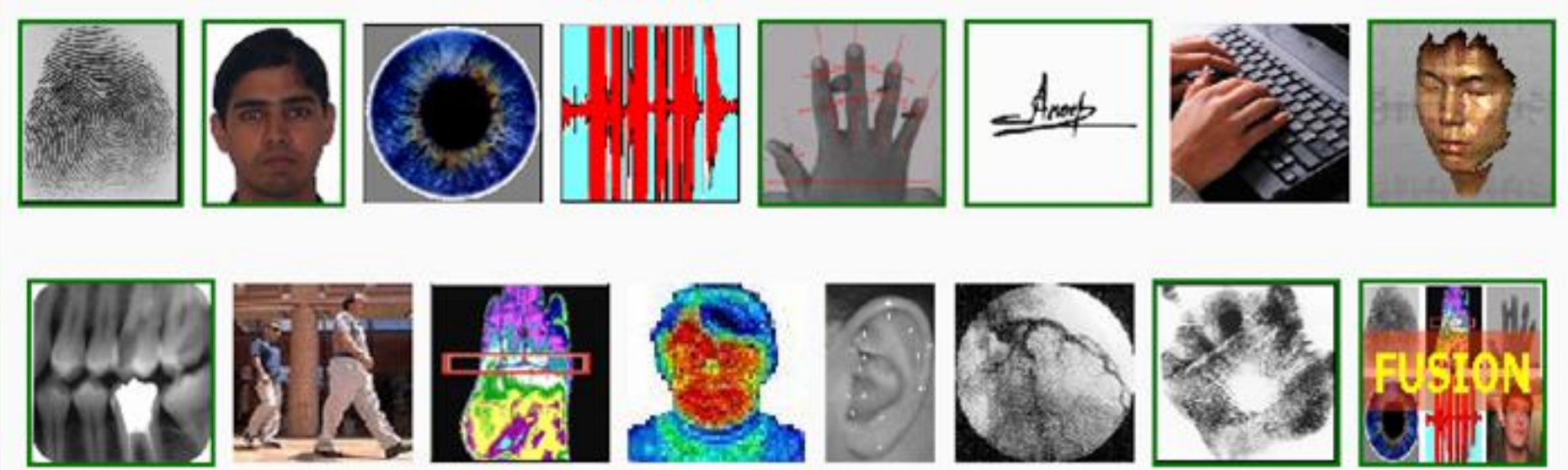
Chinese handwriting recognition

Applications of PR



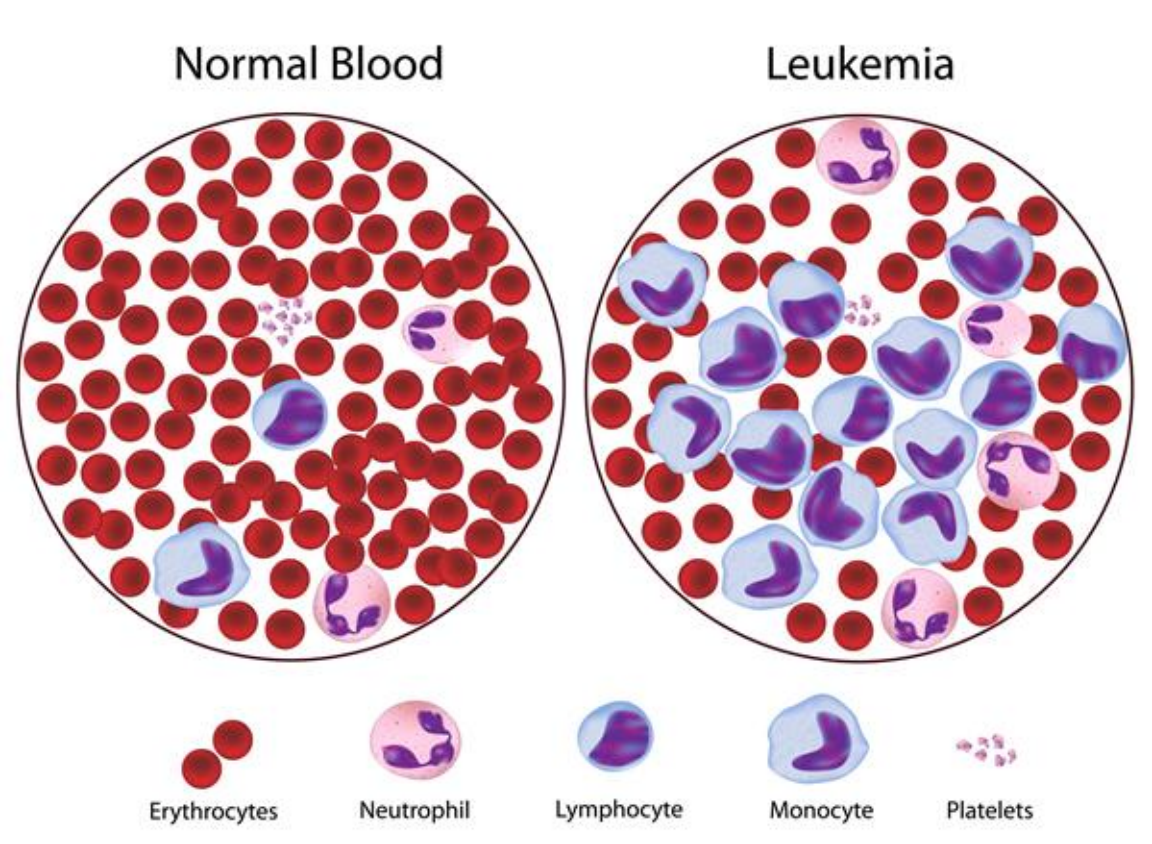
Fingerprint recognition

Applications of PR



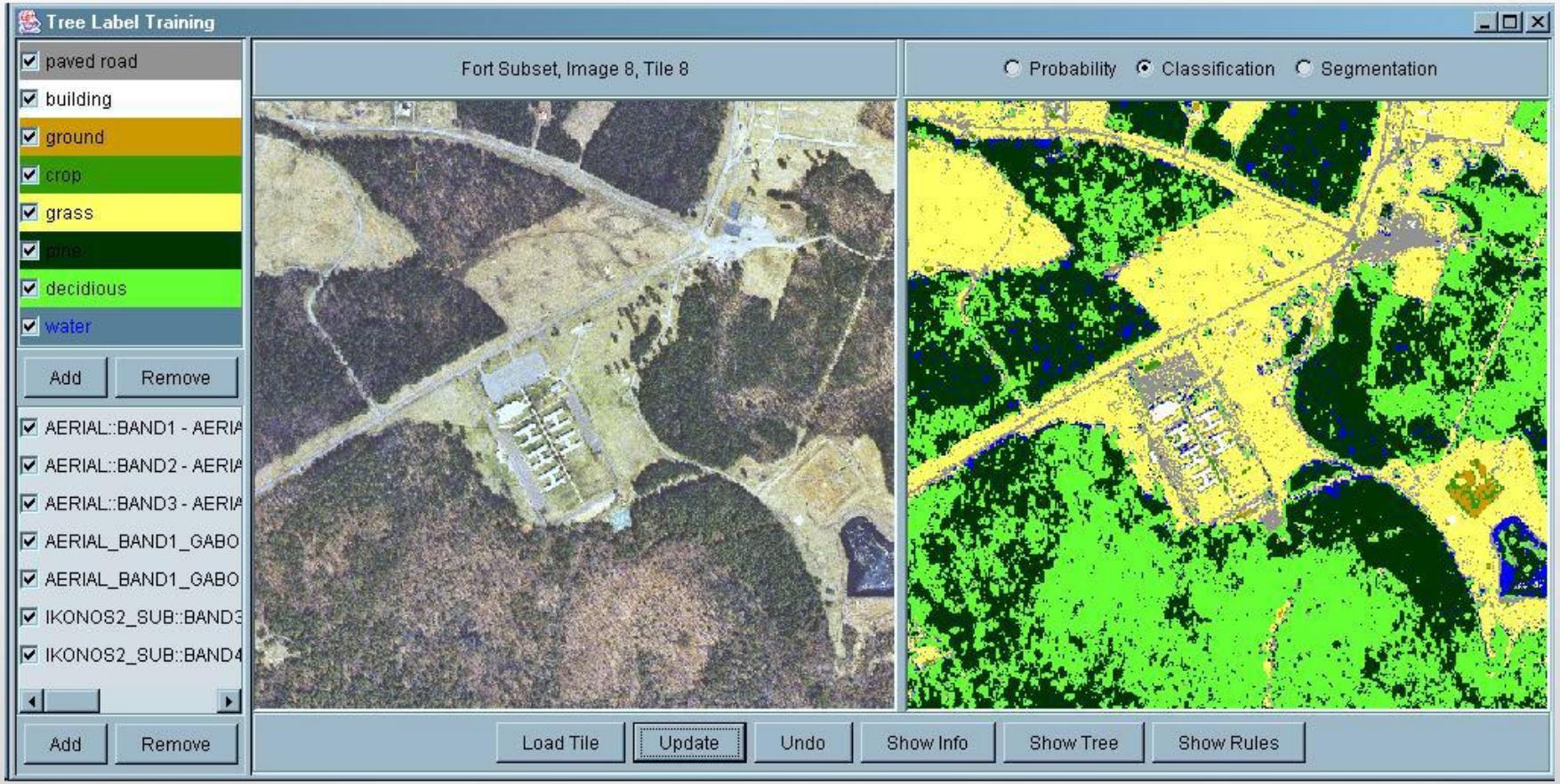
Biometric recognition.

APPLICATIONS OF PR



Classifying White Blood Cells

Applications of PR



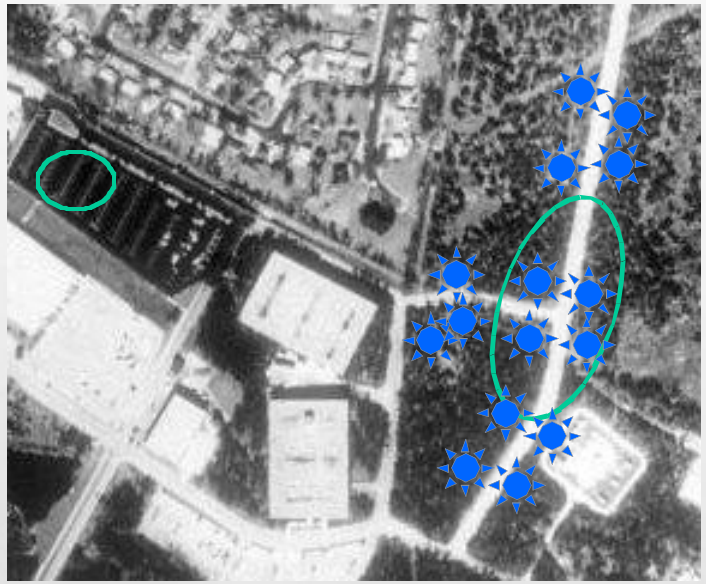
Land cover classification using satellite data

Applications of PR

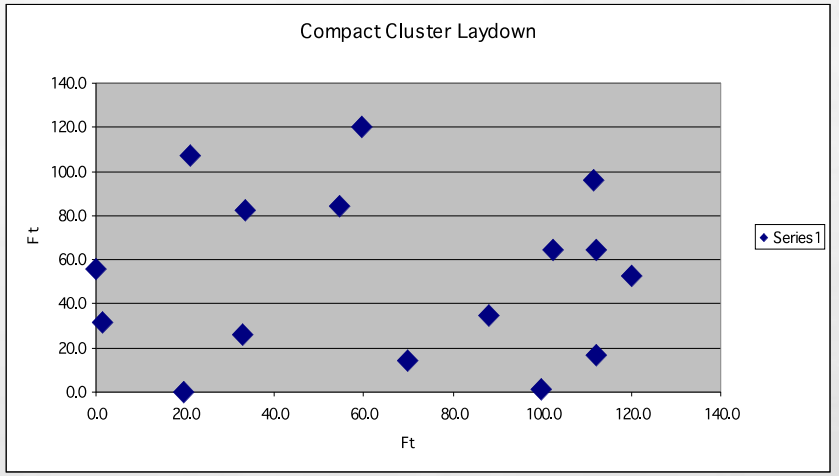


License plate recognition: US license plates.

Automatic Target Recognition







Ford 250



Harley Motocycle



Ford 350



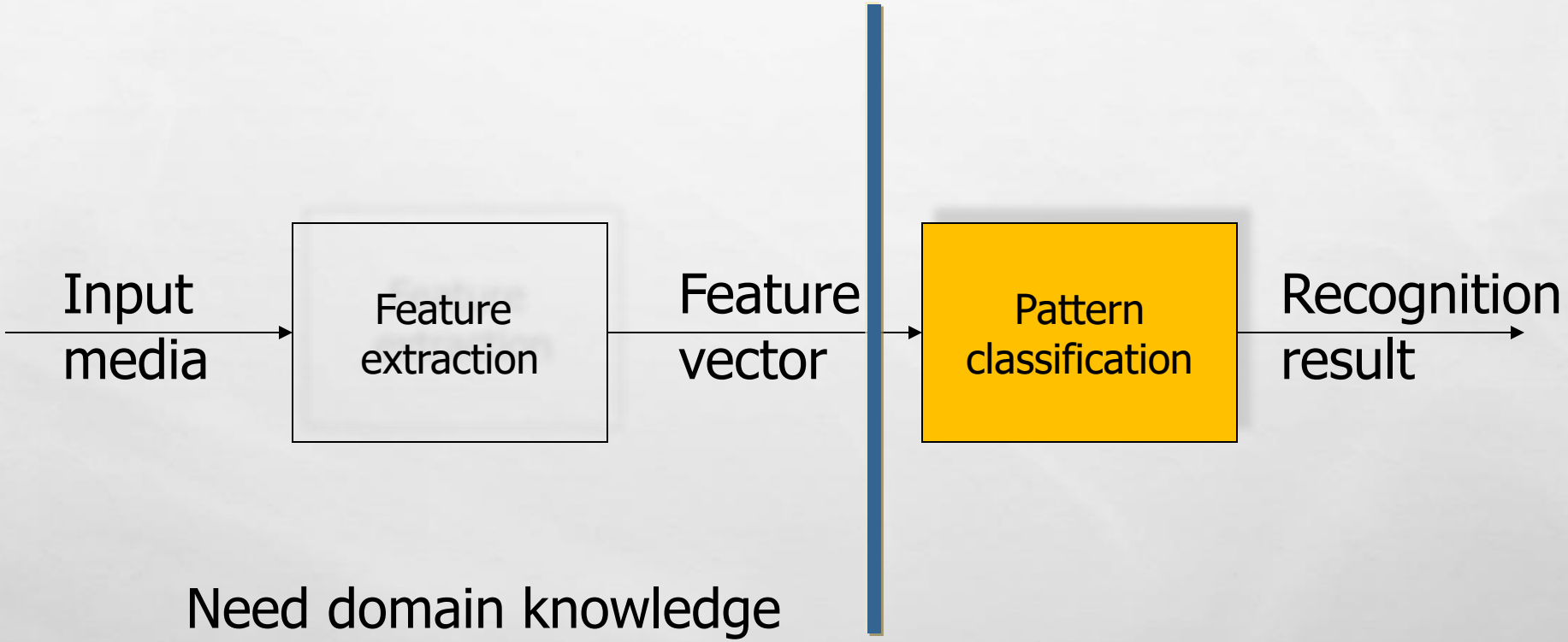
Suzuki Vitara

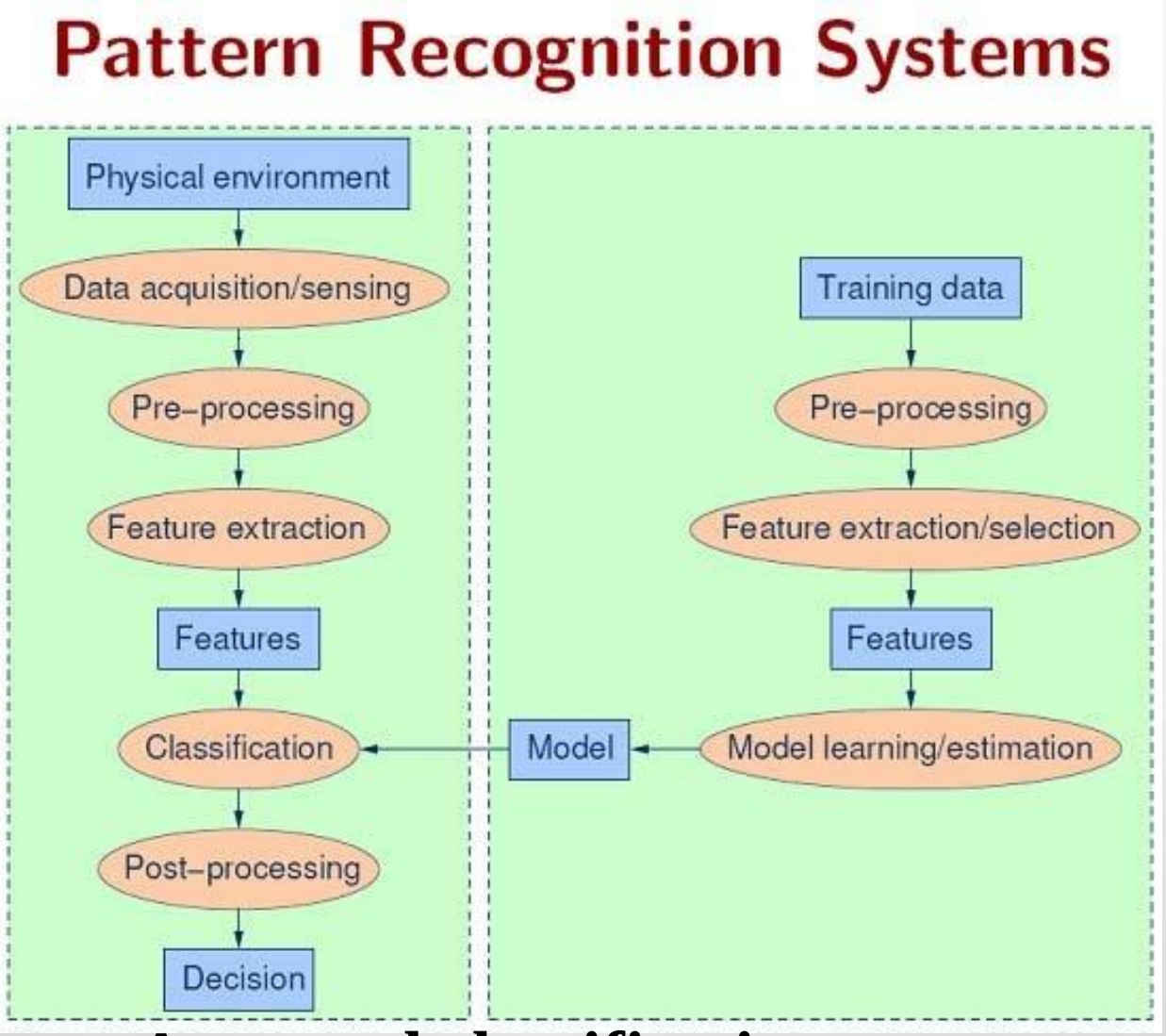


Applications of PR

- Document classification (web news classification)
  - **Input pattern**: text or html document
  - **Output classes**: semantic categories (e.g. Business, sports, …)
- Financial time series prediction
  - **Input pattern** : relation between consecutive data of time series
  - **Output values**: possible values of output (regression problem)
- Sequence analysis (bioinformatics)
  - **Input pattern** : DNA / protein sequences
  - **Output**: known types of genes
- Spam detection
  - **Input pattern** : text / image of emails
  - **Output classes**: spam / not spam

PR = Feature Extraction + Pattern Classification





A general classification process

Pattern Recognition System

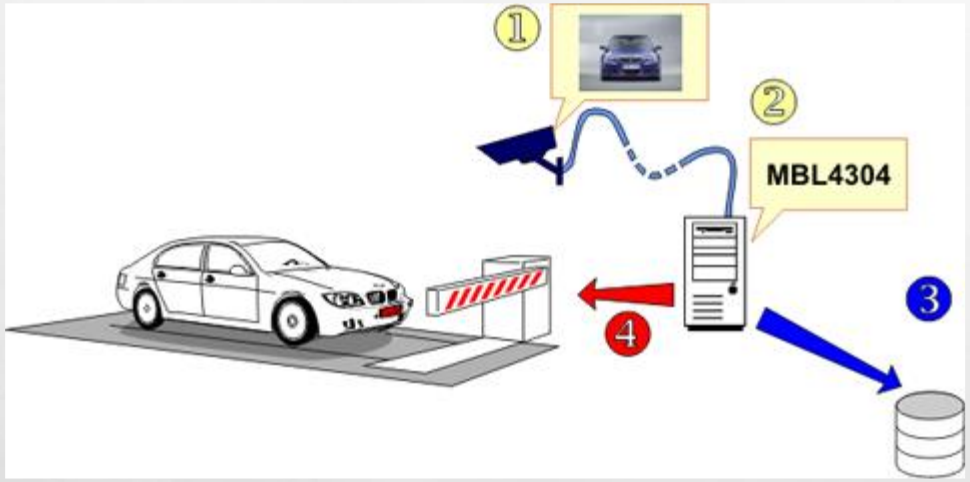
- **Data acquisition and sensing**:
  - Measurements of physical variables.
  - Important issues: bandwidth, resolution, sensitivity, distortion, latency, etc.
- **Pre-processing**:
  - Removal of noise in data.
  - Isolation of patterns of interest from the background.
- **Feature extraction**:
  - Finding a new representation in terms of features.

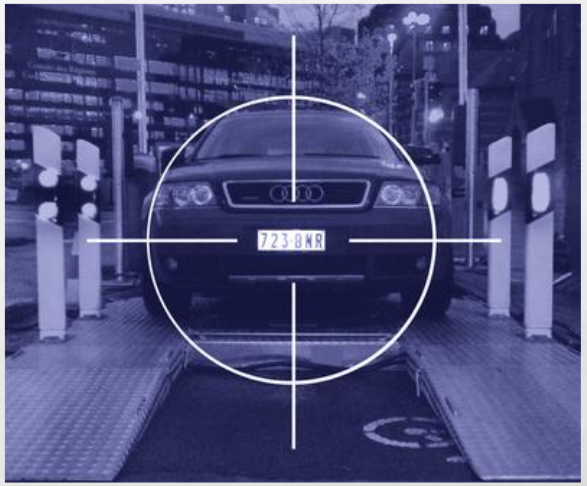
Pattern Recognition System

- **Model learning and estimation**:
  - Learning a mapping between features and pattern groups and categories.
- **Classification**:
  - Using features and learned models to **assign** a pattern to a category.
- **Post-processing**:
  - Evaluation of confidence in decisions.
  - Exploitation of context to improve performance.
  - Combination of experts.

License Plate Reading System

- Detect and read the license plates





- Modules: (i) acquisition, (ii) enhancement, (iii) segmentation, (iv) character recognition
- Accuracy, robustness & real-time

Processing Steps

- **Plate localization**: isolate the plate in image
- **Preprocessing**: plate orientation and sizing;
- **Normalization**: adjust image brightness & contrast
- **Segmentation**: find individual characters
- **Character recognition**: OCR
- **Post-processing**: rules for character placement

Challenges

- **Poor image resolution**: plate too far; low-res. Image
- **Motion blur**
- **Low contrast**: overexposure, reflection/shadows
- **Viewpoint variation and occlusion**
- **Different fonts, background**









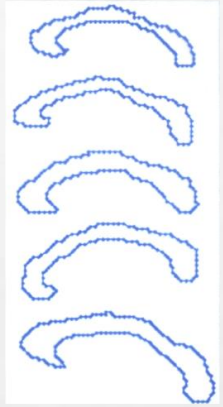


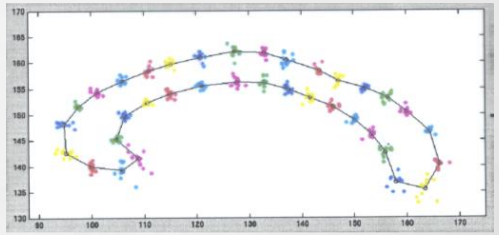


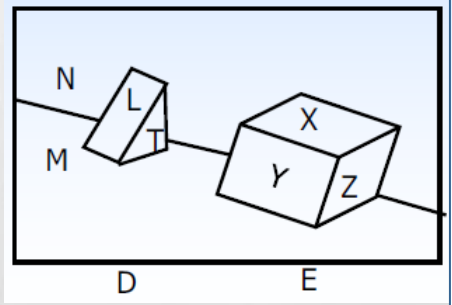


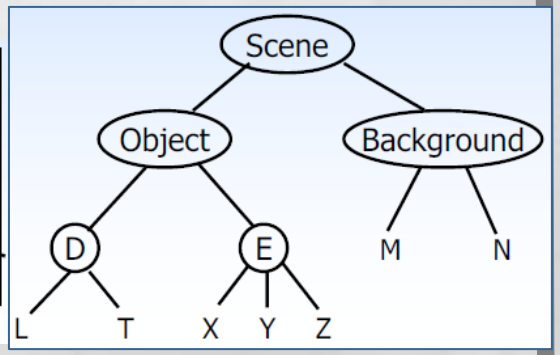
Pattern Recognition Models

- **Template matching**
  - Class-specific shape & appearance models (determine the similarity between two entities (points, curves, or shapes) of the same type)
- **Statistical (geometric)**
  - Class-specific prob. Density function (pdf)
- **Syntactic (structural)**
  - Class-specific grammar (complex objects are represented in terms of simple primitives (shapes) and their relationship; parts-based representation (represent face as eyes, mouth, nose,…))
- **Neural networks**



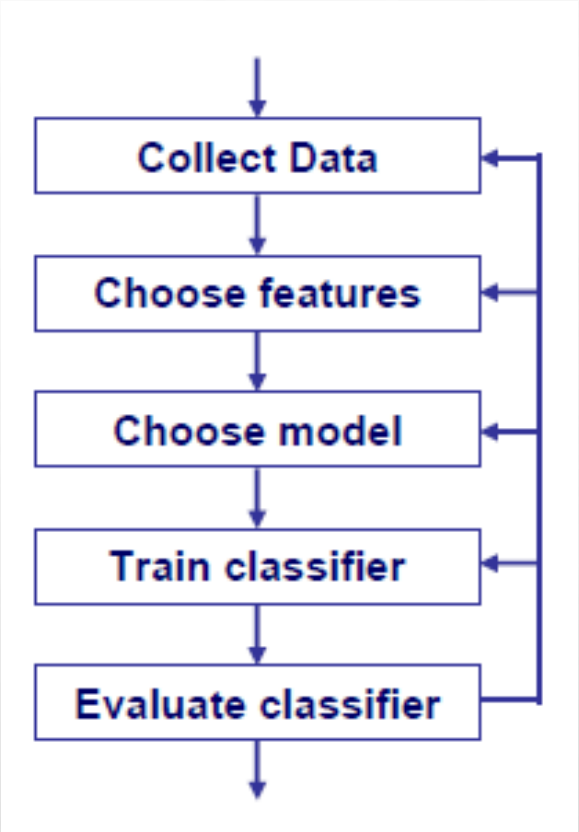






The Design Cycle

- Data collection
- Feature choice
- Model choice
- Training
- Evaluation
- Computational complexity



Assignment

- What is Data Science?
- What is deep learning?

Review Questions

- **What is the difference between classification and clustering?**
- **Mention five applications of pattern recognition?**
- **What is the difference between inter-class similarity and intra-class variability?**
- **Explain pattern recognition System steps, and prove your answer with an example.**

Questions ?

Thank You